

Humanoids at Home: Autobiographical Semantic Maps for Humanoid Motion from Memory in a Reconstructed Domestic Environment

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Fig. 1. Reconstructed environment shown with autonomous agent exploration from various views, demonstrating the agent path planning based on semantic labeling

Abstract—Semantic maps for domestic robots typically encode what an object is, rarely what it means to the people who live in the environment it navigates. We present a three-stage pipeline built on the Jefferson Drive Dataset (JDD), a high-fidelity reconstruction of a single-family home in Pittsburgh, PA, occupied by the same residents for over fifty years. First, we reconstruct the environment, covering 18 rooms across four floors, 800+ individually 3D-scanned objects, and 40+ hours of digitized 8mm/16mm film as well as archival photography, assembled as a dataset and reconstructed in high fidelity in Unreal Engine.

Second, 18 former occupants (ages 50–95) annotate the objects and spatial experiences of the reconstruction, wearing head-mounted displays and handling tangible artifacts while narrating autobiographical memories. In the semantic data gathering phase, gaze, pose, touch interactions, and transcribed speech are paired to specific objects and regions in the 3D reconstruction, producing a semantic layer in which nodes contain autobiographical and memory-based content.

Third, we drive autonomous, motion-capture-based 3D agents built from those same participants into the reconstruction. These agents navigate autonomously based on prior trajectories and goal-driven behavior derived from the memories of the space, generating scalable synthetic navigation and manipulation data grounded in real human interaction.

We report preliminary results indicating that memory-annotated regions predict motion paths in a way we propose is unique to memory-conditioned semantic mapping. In this ongoing work, we argue that memory-conditioned semantic maps offer a significant benefit to training the paths of humanoid robots operating in human homes.

I. INTRODUCTION

An autonomous robot in a human home encounters both a geometry problem and a distinctly human problem: spatial understanding is not the same as experiencing a space from the perspective of a living person. Contemporary semantic mapping has made substantial progress on the first, producing metric-semantic reconstructions [1], hierarchical scene

graphs [2], [3], and open-vocabulary language-grounded maps [4], [5].

These spatial and computer vision representations can computationally answer questions like “where is the chair.” However, they are considerably weaker on questions like “who used to sit in this chair?” or on knowing that “this chair is your grandmother’s, and since it is so fragile we don’t use it anymore.”

In personal domestic settings, object placement, use, and spatial movement are the result of decades of habit. Human judgments about where things belong are structured by episodic memory and human perception rather than explicitly by computational vision or goal-seeking behavior [6], [7].

For example, humans exhibit anticipatory gaze toward semantically predicted locations [8] and preferentially encode emotionally significant regions of space [9].

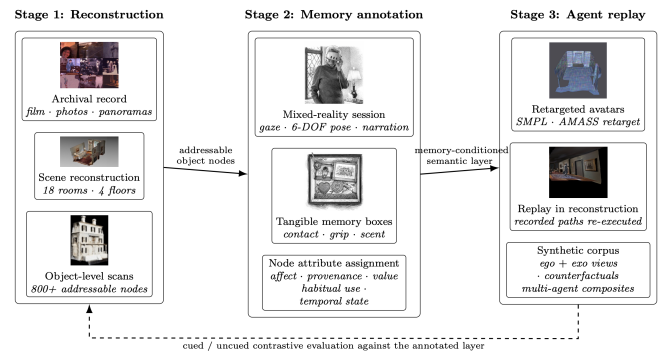


Fig. 2. System overview. Stage 1 reconstructs the residence from a longitudinal archival record, producing individually addressable object nodes. Stage 2 has former occupants annotate those nodes in a mixed-reality session, binding narration, gaze, and tangible contact events to specific objects and regions. Stage 3 retargets the same participants onto avatars and replays their recorded motion inside the annotated reconstruction to generate synthetic navigation data.

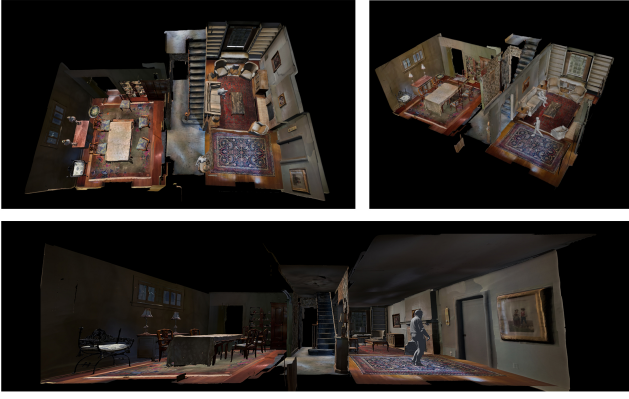


Fig. 3. Reconstructed environment shown from three perspectives, including the egocentric viewpoint occupied by the participant during annotation sessions.

A robot that models only the present configuration of a room without understanding its affective dimensions cannot accurately predict where a resident will reach, cannot infer which of two visually identical vases should be handled with care, and cannot interpret the history of a place rather than its objective appearance.

This extended abstract describes a pipeline that attaches this missing semantic layer to a 3D reconstructed environment. We discuss pilot applications which explore memory-conditioned agents to generate simulated motion paths based on memory-based behaviors in multi-agent systems. Our primary contributions are as follows:

- **A memory-annotated semantic layer.** Object and region nodes in a photorealistic reconstruction are annotated by people with decades of lived experience of the space, producing semantic labels for affective meaning, provenance (history of objects), ownership, and specific use.
- **Agent navigation in reconstructed space.** Motion-captured, 3D-scanned avatars of the same subjects are placed back into the environment, allowing recorded human trajectories to drive autonomous behaviors based on semantic labeling, later to be applied to robot egocentric views.

II. RELATED WORK

A. Semantic Mapping and Spatial Representations

Metric-semantic SLAM [1] and real-time scene graph construction [2] established hierarchical representations linking geometry to object and place labels, while open-vocabulary methods [3], [4], [5] relax the closed label set by grounding vision-language embeddings in 3D. A parallel line of research strengthens the spatial substrate of learned policies directly, injecting geometric priors from spatial foundation models [10], [11], depth experts [12], or abstracted 3D proxies [13] into vision-language-action stacks. All of these are observer-centric, in the sense that spatial semantics are derived from the sensor stream. Uniquely, our annotation layer is complementary, supplying attributes that no amount

of passive observation can recover, because the content is captured directly from the inhabitants.

B. Benchmarks and the Memory Bottleneck

Recent benchmarks do refer to memory as a limiting factor in spatial intelligence, of note for our project. For example, ESI-Bench frames spatial intelligence as a perception-action loop spanning cognitive mapping and long-term navigation [14]; OVO-S-Bench evaluates streaming egocentric reasoning and identifies allocentric mapping as the dominant failure mode, with the strongest model trailing human annotators by a wide margin [15]. Cambrian-S argues that progress requires maintaining memory across continuous experience and organizing it predictively rather than scaling context [16], and multi-view robotic evaluation shows that single-view spatial competence does not transfer to embodied settings [17]. Skill-3D introduces a scene memory that accumulates across encounters [18], though that memory is the agent’s own tool-use history rather than a human’s personal autobiographical record of the place.

We take the position that robots in a domestic environment can benefit directly from the relevant long-term memory that already exists, held by the residents, and that this memory can be semantically mapped onto the space.

C. World Models, Generative Environments, and Replay

Photorealistic reconstruction via radiance fields [19] and Gaussian splatting [20], combined with large indoor corpora [21], [22], [23] and simulators [24], underpins embodied training at scale. Recent work generates environments outright [25], [26] or translates real demonstrations into simulated counterparts for reproducible evaluation [27], and world-model policies increasingly predict future observations jointly with actions [28], [29], [30], [31], [32], a landscape now broad enough to warrant dedicated surveys [33], [34].

Controllable 4D generation extends this to dynamic scenes [35], [36], [37]. These systems deliberately trade specificity for diversity, modeling synthetic homes without data on the personal experience of being in them. We make the opposite trade: a single home, captured exhaustively with a longitudinal human record attached, in which agents are used to simulate experience of the space rather than to synthesize new spaces.

D. Human Data and Spatial Memory

Egocentric human data is an increasingly central source of data for robot learning [38], [39], [40], [41], and parametric body models [42], motion archives [43], and scene-constrained capture [44] supply the vocabulary for representing that motion. Navigation and embodied foundation models consume such data at scale [45], [46], [47], while planning-oriented work targets grounded long-horizon execution [48] and rehearsal in abstracted environments [49].

What this corpus does not contain is the experience of an agent who already knows the room. Cognitive science establishes that the movement of a lived-in inhabitant is qualitatively different from that of a neutral agent: spatial

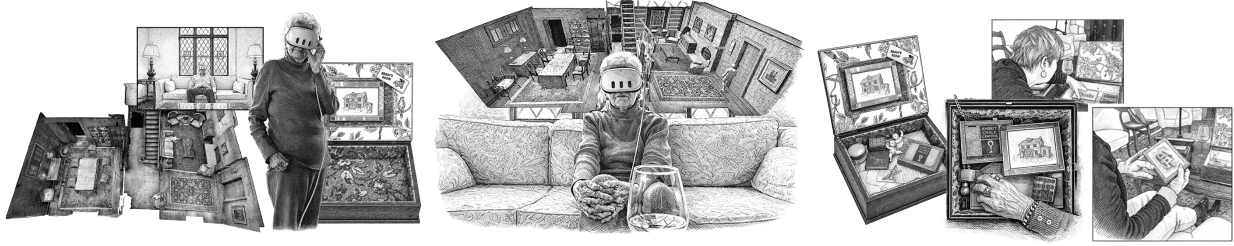


Fig. 4. AI-drawn images from the original documentation of the process of data capture for the annotation process, showing subjects exploring the space, interacting with objects, and creating annotations

TABLE I
RECONSTRUCTION AND CAPTURE SCALE.

Component	Scale	Format
3D scanned objects	800+	.obj / .fbx + textures
Rooms reconstructed	18 / 4 floors	Splat + mesh
Analog photos scanned	5,886	TIFF 600 dpi
Archival film digitized	40+ hours	MP4 24 fps
Annotation participants	18	Ages 50–95
Sensor modalities	6	See Table II
Total size	~4 TB	Mixed

TABLE II
SYNCHRONIZED ANNOTATION MODALITIES.

Modality	Rate	Signal
Egocentric video	30 fps	RGB + depth
Head pose (6-DOF)	72 Hz	Position + orientation
Gaze	90 Hz	2D gaze + pupil
Audio	44.1 kHz	8-mic field + VAD
Tangible interaction	100 Hz	Contact + grip
Narration	44.1 kHz	Transcribed speech

memory and perception share neural substrates [6], [7], episodic structure shapes navigation [50], and gaze runs ahead of the scene toward remembered locations [8], [9].

III. STAGE ONE: RECONSTRUCTION

The subject environment is a single-family residence on Jefferson Drive, Pittsburgh, PA, documented from 1965–2025 across celluloid film, analog photography, audio recordings, and high-definition conversions of 16mm and 8mm video. The original physical structure comprises 18 rooms across four floors.

a) Scene reconstruction: Each room was 3D scanned and processed through a structure-from-motion pipeline combining geometric photogrammetry with 3D Gaussian splatting [20]. Equirectangular panoramas per room provide additional coverage of specular and low-texture surfaces for lighting reconstruction.

b) Object-level capture: 800+ individual objects (furniture, heirlooms, ephemera, architectural detail) were scanned with a structured-light scanner and retopologized via an InstantMeshes-based workflow [51] into real-time-renderable meshes. Each object is a separately addressable node in its original location, which is what makes annotation at object granularity possible.

c) Temporal media registration: 40+ hours of 8mm and 16mm home film were digitized and annotated with spatial metadata binding footage to rooms and time periods. This content was used within the reconstruction to cue memories from the participants during the data gathering process.

IV. STAGE TWO: SEMANTIC AND MEMORY ANNOTATION

Eighteen participants from two closely related families, all with 30–50 years of experience of the home, annotated the reconstruction in a mixed-reality session lasting 12–28 minutes.

A. Capture Configuration

Participants wore a head-mounted display providing egocentric RGB video and 6-DOF head pose from onboard SLAM, with integrated eye tracking for gaze. A wearable microphone captured continuous narration; an instrumented set of tangible “memory boxes” (wooden containers holding original wallpaper samples, carpet fragments, retained scents, and heirloom objects) provided pressure and proximity signals when handled [52]. Table II summarizes the streams.

B. From Narration to Node Attributes

Sessions were transcribed with Claude Sonnet 4.5 (claude-sonnet-4-5-20250929), and semantic descriptors were aligned to the reconstruction. Each semantic description is linked to the node or nodes the participant was attending to when it was spoken. As a result, some or all of the following information is connected to objects and spaces in the 3D reconstruction:

- **Affective dimension:** valence and intensity of the associated recollection related to the object or space.
- **Provenance and History:** who owned or used the objects, and across which period.
- **Value:** fragility and value, distinguishing objects that are physically similar but socially not interchangeable.

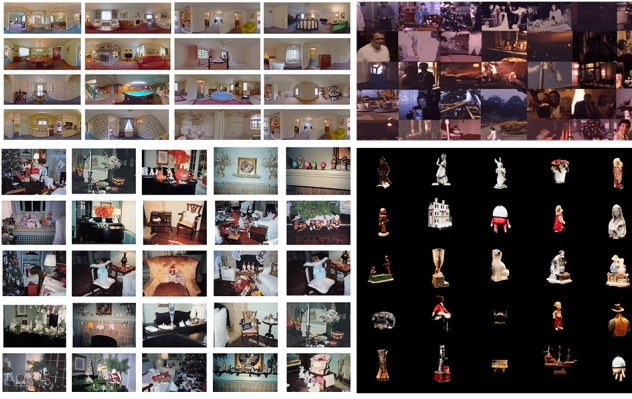


Fig. 5. Composition of the Jefferson Drive Dataset across its four capture modalities. **(top left)** Equirectangular panoramas covering 18 rooms across four floors, used as source data for scene-level reconstruction and to supplement splat coverage on specular and low-texture surfaces. **(top right)** Frames sampled from 40+ hours of digitized 8mm and 16mm home film, registered to rooms and time periods and re-projected into the reconstruction as spatially anchored surfaces. **(bottom left)** Scanned analog photographs of the same interiors, localized to rooms by background content, providing the temporal-state record of how each space was arranged across five decades. **(bottom right)** A sample of the 800+ objects captured by structured-light scanning and retopologized into individually addressable mesh nodes, the granularity at which participants assign provenance, handling class, and affective salience.

- **Habitual use:** the routine associated with a region (where people sat, which door was actually used, which chair was typically used by who).
- **Temporal state:** which configuration (in time, compared to photos or the model) of a room a memory refers to.

These attributes are not observable from a sensor placed in the room today. A robot may be able to perceive that a vase is ceramic but cannot remember the nostalgia attached to it as a wedding gift from a lost loved one.

V. STAGE THREE: AGENT REPLAY IN SIMULATION

The final stage, the semantic annotation stage, yields sparse, human-generated data: eighteen sessions, each under half an hour.

Each participant was 3D scanned and rigged to a parametric body model [42], and their recorded head pose, gaze, and contact events were retargeted onto that avatar [43], [44].

The avatar is then instantiated inside the reconstruction and the movement paths from their memories are re-executed, which decouples the recorded motion from the recording viewpoint. Because the motion originates in a real person moving through a real space they remember, the resulting synthetic corpus retains subtleties of cognitive experience that generated or randomly sampled navigation trajectories lack [25], [26].

VI. PRELIMINARY FINDINGS

Although analysis is ongoing, we report three preliminary observations.

Memory-annotated regions attract disproportionate attention. Consistent with prior findings, we predict that gaze and movement concentrate on regions corresponding to verbally recalled content [8], [9]. Participants describing holiday gatherings fixated densely around the dining table; participants recalling basement activity directed gaze toward stairway affordances and lower-floor doorways well before approaching them [50].

Movement precedes object visibility. Contact and gesture data show participants orienting toward remembered object locations, including cases where the object existed only in the virtual layer with no physical correlate [7], [6]. This is consistent with what a domestic robot needs in order to predict where a resident expects to find something.

Cross-modal cues affect focus. Participants who handled the olfactory and textile artifacts showed longer focus durations in the virtual room than those who did not [9], suggesting that non-visual channels meaningfully modulate spatial behavior.

VII. IMPLICATIONS FOR SEMANTIC-AWARE NAVIGATION

Meaning Based Manipulation Two objects with identical geometry and identical open-vocabulary class may warrant different grasps or uses. Current open-vocabulary maps [3], [5] have no channel for this distinction.

Habitual Use and Object Location Habitual-use annotations give a distribution over where a resident believes an object is, which may diverge from where it currently is. For assistive robots in the homes of older adults, resolving that divergence is a frequent and consequential task.

Grounding History-Referential Language Instructions in long-inhabited homes routinely reference provenance and period rather than appearance. Temporal-state annotations paired with registered archival media provide supervision for grounding such references, complementing spatially grounded planning benchmarks [48].

VIII. CONCLUSION

We have described a pipeline for a work-in-progress sim2real system that reconstructs a long-inhabited home, has its former occupants annotate the reconstruction with memory and meaning at object and space level, and then deployed 3D-scanned, motion-captured agents through the annotated space to generate simulated navigation data grounded in real human movement.

The premise is that the semantics an autonomous humanoid robot needs to navigate a home are partly unobservable from the home itself and must be elicited from the people who lived in it.

We hope the resulting corpus is useful to the mapping and navigation community as a testbed for representations that encode not only what is where, but what it means and to whom.

REFERENCES

- [1] A. Rosinol, M. Abate, Y. Chang, and L. Carlone, “Kimera: an open-source library for real-time metric-semantic localization and mapping,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020.
- [2] N. Hughes, Y. Chang, and L. Carlone, “Hydra: A real-time spatial perception system for 3D scene graph construction and optimization,” in *Robotics: Science and Systems (RSS)*, 2022.
- [3] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula *et al.*, “ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024.
- [4] C. Huang, O. Mees, A. Zeng, and W. Burgard, “Visual language maps for robot navigation,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023.
- [5] N. M. M. Shafiullah, C. Paxton, L. Pinto, S. Chintala, and A. Szlam, “CLIP-Fields: Weakly supervised semantic fields for robotic memory,” in *Robotics: Science and Systems (RSS)*, 2023.
- [6] J. O’Keefe and L. Nadel, *The Hippocampus as a Cognitive Map*. Oxford: Clarendon Press, 1978.
- [7] P. Byrne, S. Becker, and N. Burgess, “Remembering the past and imagining the future: A neural model of spatial memory and imagery,” *Psychological Review*, vol. 114, no. 2, pp. 340–375, 2007.
- [8] B. W. Tatler, M. M. Hayhoe, M. F. Land, and D. H. Ballard, “Eye guidance in natural vision: Reinterpreting salience,” *Journal of Vision*, vol. 11, no. 5, 2011.
- [9] J. L. McGaugh, “Memory: A century of consolidation,” *Science*, vol. 287, no. 5451, pp. 248–251, 2000.
- [10] H. Peng, H. Li, Y. Dai, Y. Lan, Y. Luo, T. Qi, Z. Zhang, Y. Zhan, J. Zhang, W. Xu, and Z. Liu, “OmniVGGT: Omni-modality driven visual geometry grounded transformer,” *arXiv preprint*, 2026.
- [11] Z. Zhang, H. Li, Y. Dai, Z. Zhu, L. Zhou, C. Liu, D. Wang, F. E. H. Tay, S. Chen, Z. Liu, Y. Liu, X. Li, and P. Zhou, “From spatial to actions: Grounding vision-language-action model in spatial foundation priors,” *arXiv preprint*, 2026.
- [12] T. Yuan, Y. Liu, C. Lu, Z. Chen, T. Jiang, and H. Zhao, “DepthVLA: Enhancing vision-language-action models with depth-aware spatial reasoning,” *arXiv preprint*, 2026.
- [13] Y. Liu, F. Zhan, K. Zhou, Y. Du, P. P. Liang, and H. Pfister, “Abstract 3D perception for spatial intelligence in vision-language models,” *arXiv preprint arXiv:2511.10946*, 2026.
- [14] Y. Hong, J. Liu, H. Yin, M. Li, L. Guibas, L. Fei-Fei, J. Wu, and Y. Choi, “ESI-Bench: Towards embodied spatial intelligence that closes the perception-action loop,” *arXiv preprint*, 2026.
- [15] Y. Li, P. Liu, Y. Zang, Z. Shi, Q. Fu, H. Hao, and J. Lu, “OVO-S-Bench: A hierarchical benchmark for streaming spatial intelligence in multimodal LLMs,” *arXiv preprint arXiv:2606.03890*, 2026.
- [16] S. Yang, J. Yang, P. Huang, E. Brown, Z. Yang, Y. Yu, S. Tong, Z. Zheng, Y. Xu, M. Wang, D. Lu, R. Fergus, Y. LeCun, L. Fei-Fei, and S. Xie, “Cambrian-S: Towards spatial supersensing in video,” *arXiv preprint arXiv:2511.04670*, 2025.
- [17] Z. Feng, Z. Kang, Q. Wang, Z. Du, J. Yan, S. Shi, C. Yuan, H. Liang, Y. Deng, Q. Li *et al.*, “Seeing across views: Benchmarking spatial reasoning of vision-language models in robotic scenes,” *arXiv preprint*, 2026.
- [18] H. Li, Z. Hu, J. Wang, H. Fan, and Y. Yang, “Skill-3D: Evolving scene-aware skills for agentic 3D spatial reasoning,” *arXiv preprint arXiv:2606.07456*, 2026.
- [19] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, “NeRF: Representing scenes as neural radiance fields for view synthesis,” in *Proc. European Conf. Computer Vision (ECCV)*, 2020.
- [20] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graphics*, vol. 42, no. 4, 2023.
- [21] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, “ScanNet: Richly-annotated 3D reconstructions of indoor scenes,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [22] A. Chang, A. Dai, T. Funkhouser, M. Halber, M. Nießner, M. Savva, S. Song, A. Zeng, and Y. Zhang, “Matterport3D: Learning from RGB-D data in indoor environments,” in *Proc. Int. Conf. 3D Vision (3DV)*, 2017.
- [23] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang *et al.*, “Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI,” in *NeurIPS Datasets and Benchmarks Track*, 2021.
- [24] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra, “Habitat: A platform for embodied AI research,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2019.
- [25] X. Wang, L. Liu, Y. Cao, R. Wu, W. Qin, D. Wang, W. Sui, and Z. Su, “EmbodiedGen: Towards a generative 3D world engine for embodied intelligence,” *arXiv preprint arXiv:2506.10600*, 2025.
- [26] A. Choi, X. Wang, Z. Su, and W. Xu, “Scaling sim-to-real reinforcement learning for robot VLAs with generative 3D worlds,” *arXiv preprint*, 2026.
- [27] Y. Jangir, Y. Zhang, P.-C. Lo, K. Yamazaki, C. Zhang, K.-H. Tu, T.-W. Ke, L. Ke, Y. Bisk, and K. Fragkiadaki, “RobotArena ∞ : Scalable robot benchmarking via real-to-sim translation,” *arXiv preprint*, 2026.
- [28] M. J. Kim, Y. Gao, T.-Y. Lin, Y.-C. Lin, Y. Ge, G. Lam, P. Liang, S. Song, M.-Y. Liu, C. Finn, and J. Gu, “Cosmos policy: Fine-tuning video models for visuomotor control and planning,” *arXiv preprint*, 2026.
- [29] S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, Y. Zhu, L. Fan, and J. Jang, “World action models are zero-shot policies,” *arXiv preprint arXiv:2602.15922*, 2026.
- [30] W. Huang, Y.-W. Chao, A. Mousavian, M.-Y. Liu, D. Fox, K. Mo, and L. Fei-Fei, “PointWorld: Scaling 3D world models for in-the-wild robotic manipulation,” *arXiv preprint arXiv:2601.03782*, 2026.
- [31] J. Sun, W. Zhang, Z. Qi, S. Ren, Z. Liu, H. Zhu, G. Sun, X. Jin, and Z. Chen, “VLA-JEPA: Enhancing vision-language-action model with latent world model,” *arXiv preprint*, 2026.
- [32] Z. Wang, K. Sivakumar, J. Shang, Y. Hu, Z. Xie, R. Gong, X. Zhang, and K. Schmeckpeper, “Efficient sim-to-real transfer of world-action models from synthetic priors,” *arXiv preprint*, 2026.
- [33] F. Wang, Z. Wang, G. Pei, M. Zhang, C. Liang, J. Hu, Z. Li, J. Wu, N. Han, Z. Zhang, J. Qi, H. Wu, S. Zhang, P. Zheng, J. Pan, D. Navarro-Alarcon, S. Liu, and P. Zhou, “World models for robotic manipulation: A survey,” *arXiv preprint arXiv:2606.00113*, 2026.
- [34] L. Kong, Y. Yang, J. Mei, Y. Liu, A. Liang, D. Zhu, D. Lu, W. Yin, X. Hu, M. Jia, J. Deng, K. Zhang, Y. Wu, T. Yan, S. Gao, S. Wang, L. Li, L. Pan, Y. Liu, J. Zhu, W. T. Ooi, S. C. H. Hoi, and Z. Liu, “3D and 4D world modeling: A survey,” *arXiv preprint arXiv:2509.07996*, 2025.
- [35] S. Zheng, M. Yin, W. Hu, X. Li, Y. Shan, and Y. Fu, “VerseCrafter: Dynamic realistic video world model with 4D geometric control,” *arXiv preprint*, 2026.
- [36] H. Mun, I.-H. Jin, S. Kim, and K. Kong, “LivingWorld: Interactive 4D world generation with environmental dynamics,” *arXiv preprint*, 2026.
- [37] H. Wang, H. Zhou, T. Gu, and L. Yan, “ST-Gen4D: Embedding 4D spatiotemporal cognition into world model for 4D generation,” *arXiv preprint*, 2026.
- [38] K. Grauman, A. Westbury, E. Byrne *et al.*, “Ego4D: Around the world in 3,000 hours of egocentric video,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [39] K. Grauman, A. Westbury, L. Torresani *et al.*, “Ego-Exo4D: Understanding skilled human activity from first- and third-person perspectives,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [40] R. Punamiya, S. Kareer, Z. Liu, J. Citron, R.-Z. Qiu, X. Cai, A. Gavryushin, J. Chen, D. Liconti, L. Y. Zhu, S. Song, J. Hoffman, and D. Xu, “EgoVerse: An egocentric human dataset for robot learning from around the world,” *arXiv preprint arXiv:2604.07607*, 2026.
- [41] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, “Universal manipulation interface: In-the-wild robot

- teaching without in-the-wild robots,” in *Robotics: Science and Systems (RSS)*, 2024.
- [42] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, “SMPL: A skinned multi-person linear model,” *ACM Trans. Graphics*, vol. 34, no. 6, 2015.
 - [43] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, “AMASS: Archive of motion capture as surface shapes,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2019.
 - [44] M. Hassan, V. Choutas, D. Tzionas, and M. J. Black, “Resolving 3D human pose ambiguities with 3D scene constraints,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2019.
 - [45] J. Zhang, A. Li, Y. Qi, M. Li, J. Liu, S. Wang, H. Liu, G. Zhou, Y. Wu, X. Li, Y. Fan, W. Li, Z. Chen, F. Gao, Q. Wu, Z. Zhang, and H. Wang, “Embodied navigation foundation model,” *arXiv preprint*, 2025.
 - [46] ACE-Brain Team, “ACE-Brain-0.5: A unified embodied foundational model for physical agentic AI,” ACE Robotics, Tech. Rep., 2026.
 - [47] H. Fang, J. Duan, D. Clay, S. Wang, S. Liu, W. Huang, X. Fan, W.-C. Tsai, S. Chen, Y. R. Wang, S. Xing, J. Cho, J. S. Park, A. Eftekhari, P. Sushko, J. Chai, Z. Ren, A. Farhadi, D. Fox, and R. Krishna, “MolmoAct2: Action reasoning models for real-world deployment,” *arXiv preprint arXiv:2605.02881*, 2026.
 - [48] S. Jung, H. Song, D.-H. Kim, R. Tan, J. Gao, Y. J. Lee, and D. Kim, “Spatially grounded long-horizon task planning in the wild,” *arXiv preprint*, 2026.
 - [49] Z. Shen, J. Du, Z. Guo, H. Luo, L. Peng, J. T. Zhou, H. Luo, and T. Li, “Plan in sandbox, navigate in open worlds: Learning physics-grounded abstracted experience for embodied navigation,” *arXiv preprint*, 2026.
 - [50] N. K. Speer, J. M. Zacks, and J. R. Reynolds, “Human brain activity time-locked to narrative event boundaries,” *Psychological Science*, vol. 18, no. 5, pp. 449–455, 2007.
 - [51] W. Jakob, M. Tarini, D. Panozzo, and O. Sorkine-Hornung, “Instant field-aligned meshes,” *ACM Trans. Graphics*, vol. 34, no. 6, 2015.
 - [52] H. Ishii and B. Ullmer, “Tangible bits: Towards seamless interfaces between people, bits and atoms,” in *Proc. ACM CHI Conference on Human Factors in Computing Systems*, 1997.